

The Line-Following Robot

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About the Robot:

Course Conditions:

- Defining the Line:
 - Width: 1.8 – 2.5 cm
 - Line should contrast with the floor
 - Line and floor should be of contrasting colours (white on black)
- Lighting:
 - Indoor lighting conditions
 - No heavy shadows -> for simplicity
- Surface:
 - Smooth flat surface
 - No obstacles, doors, animals

Picking Line marking material:

Why should I choose this?	Why should I not choose this?	Material
• Easily control width of the line • Wide variety of colours • Durable	• Time consuming – time to dry • Changes may be permanent	Paint
• Variety of colours available • Works well with paper	• Can be spotty, faded – broken lines can cause the robot to go off track	Marker Pen
• Able to create, adjust, and remove any unwanted tape – change/edit the track easily	• Glossy dark tape reflects more light, so the robot may think there is no	Tape Lines

• Easy to obtain desired widths, and type of surfaces (e.g. glossy or matt) • Glossy white or matt black provides the highest contrast	line and eventually go off track	
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I chose to use the tape as a material of choice for my track (line) as in my opinion it will give me the best ‘bang for the buck’. This is because for the same amount of price of a sharpie for example or even less I get more material that can be easily cut to the desired size, provide higher contrasts and result in a continuous line. Therefore, I chose to use tape as my choice of material.

Which tape?

According to my research blue low-adhesive tape is the best because it is not quite light nor dark. Therefore, if this tape is placed in a dark background, then the tape seems light and when stuck on light background it appears to be dark, therefore provides the best of both worlds – making it more usable than any other colour. Furthermore, low adhesion means that when it is peeled off there is going to be little to no residue left on the surface it was stuck onto. This is also important as I would save lots of time not cleaning up the residue and helps me keep my surroundings tidy.

However, blue reflects less infrared light so it would be a bad choice if the robot is using IR sensors in the line detecting sensing circuit. In this project I am going to be using light dependant resistors instead of IR sensors to keep the overall cost down and save space.

The track and the Line:

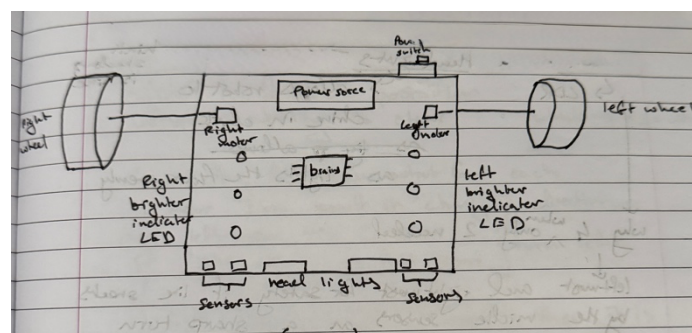
For simplicity’s sake in the initial stages of testing, in the track I am not going to make turns more than 22 degrees.

The lines need to always be continuous.

For tracks which have ‘X’ or ‘Y’ crossings my robot will pick either path and will not go off track.

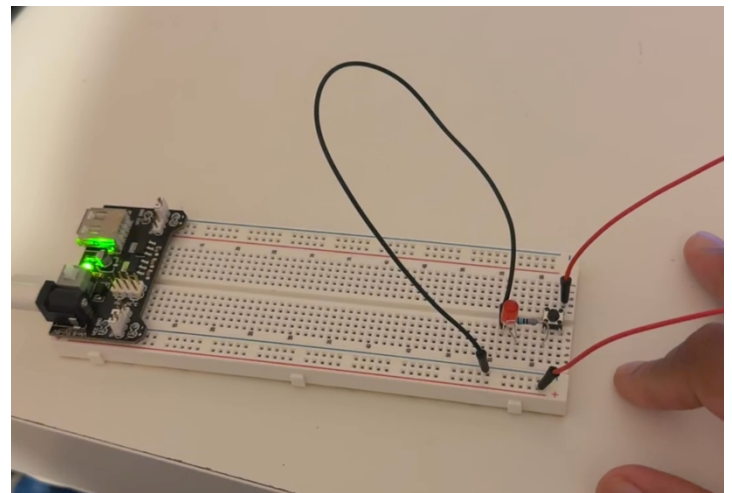
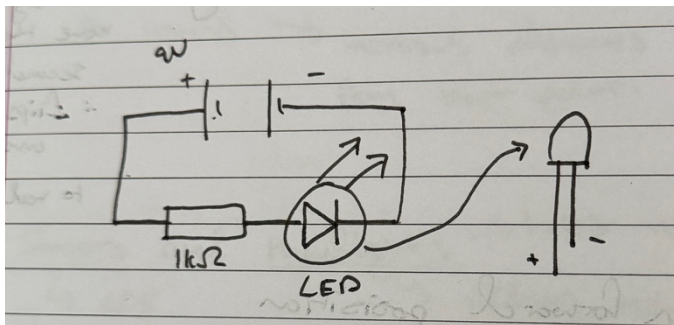
Size of the body:

Again, for simplicity I am going to be using a lunchbox to be the body of my robot and the internal arrangement of the electronics will be decided based on the size of lunchbox I get. Below is a diagram of the layout of the robot I want to make.



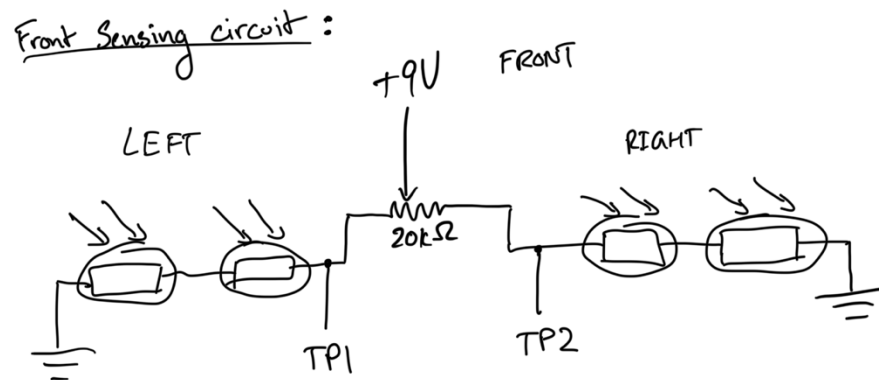
Power:

To supply power to the electrical components I am going to be using a 9V battery as it can be found easily in supermarkets and not requiring a connection from the mains – to increase distances the robot can travel and make it easier for me to test make changes in the robot as the mains wire will not interfere.



The image on the left is the circuit diagram for the power indicator – showing if power is getting supplied to the components or not. The image on the right shows the circuit I have made. As in this point of time I didn't have 9V batteries I used the power supply module from my Arduino kit and supplied 5V to the breadboard.

Sensing Circuit:



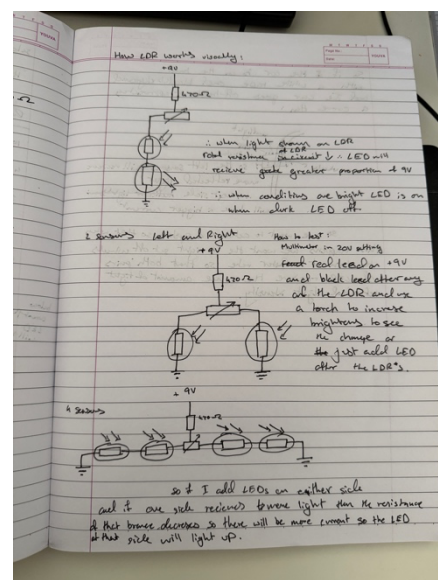
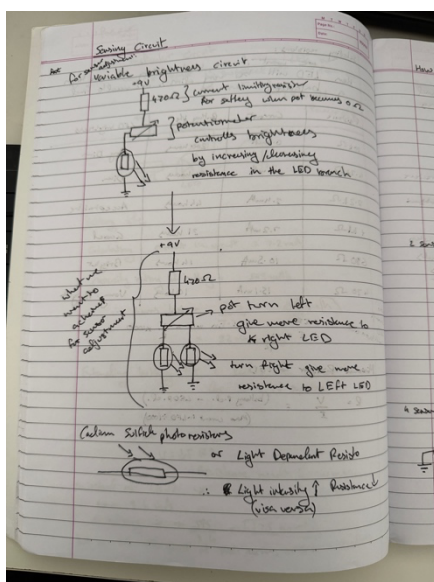
Explanation of the Circuit: I am going to supply 9V to a 20kΩ potentiometer. Under ideal conditions, the potentiometer will be set to divide the total resistance equally, with 10kΩ on each side. This will divide the 9V supply equally, resulting in a potential difference (p.d.) of 4.5V across each side.

When the robot is positioned in the centre of the track, with a white background and a black line, both LDRs should receive the same amount of reflected light from the headlight. Therefore, both sides of the circuit should produce approximately 4.5V.

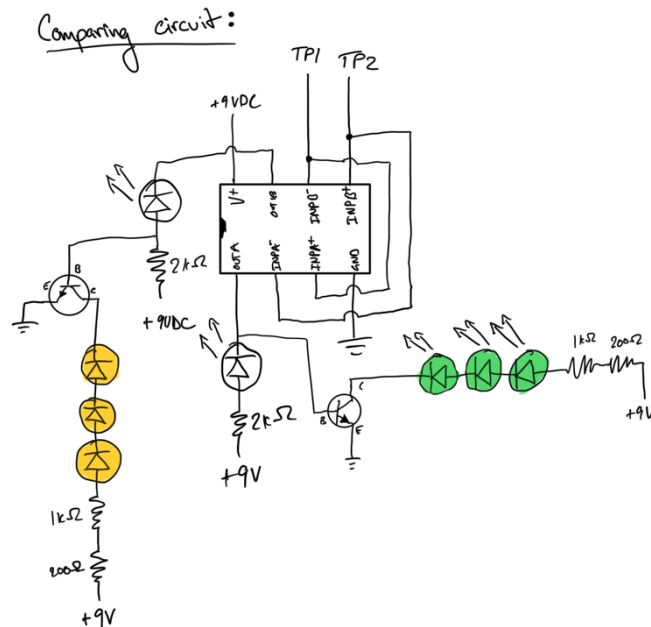
However, if one side of the robot moves over the black line, the LDR on that side will receive less reflected light because the black line absorbs more of the light from the headlight. As the light intensity decreases, the resistance of the LDR increases, causing a greater p.d. across that branch of the circuit.

I will use these changes in voltage as inputs to my comparator IC (indicated by the TP1 and TP2 points on the diagram), allowing the circuit to detect when the robot moves away from the centre of the track.

The following images are pictures of my notes that shows the step by step progress that led me to the sensing circuit stated above.



Comparing Circuit/brains:



Circuit explanation: The comparator IC I am using is the **LM393**, which contains two independent comparators, allowing it to compare two pairs of input voltages simultaneously. I am using this dual-comparator function so that each comparator output can be used to control one side of the robot's motors.

For example, if the right-hand LDRs move over the black line, they will receive less reflected light. This causes their resistance to increase, resulting in a greater p.d. across the right branch of the sensing circuit. Therefore, TP2 will have a higher voltage than TP1.

In the comparator circuit shown above, this causes **output A to be activated**, because the voltage at the inverting (-) input is greater than the voltage at the non-inverting (+) input. When output A is activated, it provides a path to ground, completing the circuit connected to this output.

As a result, the LED connected to output A will illuminate. Current then flows through this LED and into the base of the NPN transistor, causing the transistor to switch on. This allows current to flow through the three green LEDs, which represent the left-side motors.

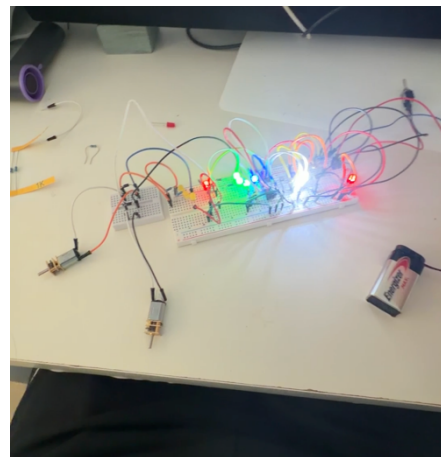
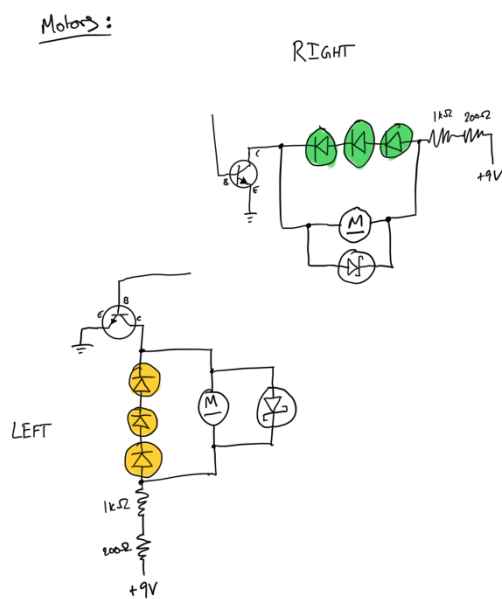
The left-side motors will therefore run, causing the robot to turn back towards the centre of the track and correct its position. This means the circuit uses the difference

between TP1 and TP2 to determine which side of the robot needs to be driven in order to keep the robot following the line.

However, using PNP transistors would have been simpler because it would not require the additional LED and could have allowed me to connect the three motor-indicator LEDs directly to the transistor, with the same transistor controlled by output A. This is because an NPN transistor requires current to flow into its base in order to switch on. The LM393 comparator has an open-collector output, meaning that it can sink current to ground but cannot supply current from its output. Therefore, I could not directly use the comparator output to provide the base current required by the NPN transistor.

I chose to use NPN transistors because I already had several available from my Arduino kit. Rather than replacing them with PNP transistors, I designed the additional LED stage as a way of providing the required current to the base of the NPN transistor. This allowed me to use the components I already had while still achieving the required switching function.

Adding the Motors:



Circuit explanation: I have added 12V DC gear motors that are reduced to 500 RPM in parallel to the 3 LEDs and I have also added a Schottky Barrier Diode parallel to the motor which is reverse bias so when the motor is accidentally turned in the wrong direction no current flows through the rest of the circuit damaging other components.

The reason why I used a transistor allowing a greater current to flow through the LEDs is because adding such a motor would require a greater current to be able to turn on.

The image on the right shows how the motors connected with the breadboard and other electrical components.

The Body:

Wheels:

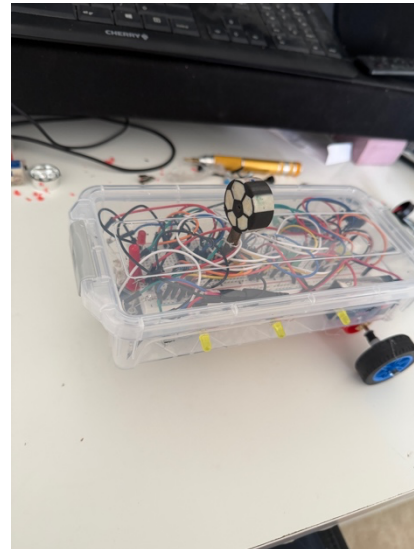


The image on the left shows the set of wheels I used for the robot's movement. I chose Lego wheels because they were the most suitable option available to me. They have rubber tyres, which provide sufficient grip for the robot to move effectively on smooth surfaces without excessive slipping. They were also readily available to me, as I already had several Lego sets.

The image on the right shows how I connected the wheel to the motor. I used Henkel Loctite superglue to permanently attach the wheel to the motor shaft. The Lego piece, which was covered in electrical tape, was filled with superglue before the motor's metal shaft was inserted into it. I then held the wheel in the correct position for approximately 30 minutes, ensuring that the wheel remained perpendicular to the motor shaft while the glue set. After this, the wheel was securely attached to the motor shaft.

Although this method worked effectively, it has the disadvantage of making the wheel and motor difficult to separate or replace. A custom-made coupler would have been a better solution, as it would have allowed the wheel and motor to be easily removed and replaced while also providing a more secure and precise connection.

V1:



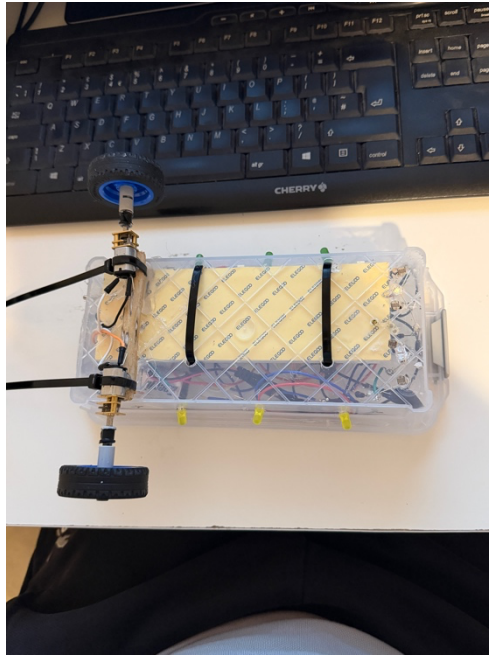
For the main body of the robot, I used a lunchbox that I purchased from Tesco for £1.50. I had several different options available for the body, but I chose this lunchbox because it was long enough to accommodate the breadboard and the other components. I considered designing a custom PCB, which would have allowed me to use a smaller and more compact enclosure. However, as this robot was being developed as a prototype rather than a finished product, I decided that designing a PCB would be unnecessarily time-consuming at this stage. Using a breadboard also made it much easier to modify and troubleshoot the circuit during development.

I decided to position the motors and wheels underneath the lunchbox rather than inside it. If the motors were placed inside the lunchbox, the breadboard would have needed to be mounted higher to provide enough space for them. This would have made the robot less practical to work on and would have made diagnosing problems with the motors more difficult, as I would have needed to remove the breadboard and use longer wires to access the motors. Therefore, mounting the motors and wheels underneath the lunchbox was a more practical solution.

For the motor and wheel assembly, I used the body of a small torch as a supporting structure. I secured the motors in place using zip ties. However, this created a significant problem because of the way the zip ties were positioned. The motors could easily be pushed further into the body, causing their alignment to change. This meant that the wheels were not always positioned correctly, which affected the robot's movement. This became the biggest mechanical issue during the build and was one of the main reasons I decided to produce a second version of the robot.

I also added a caster wheel at the front of the robot. The purpose of this wheel was to support the front of the robot, prevent the sensors from rubbing against the surface, and allow the robot to turn more easily. I obtained the caster wheel by removing it from an old, unused travel bag. However, this modification was also unsuccessful. The wheel was too loose and was not securely connected to its mounting point. As a result, when the robot was switched on, the loose caster wheel caused the robot to become unstable and fall off the track. This highlighted the importance of having a secure and rigid mechanical structure, which I addressed when developing the second version.

V2:



The image on the left shows that I removed the caster wheel and redesigned the movement module to improve the stability and alignment of the robot. For the new design, I cut a bamboo stick to the same length as the width of the lunchbox and secured it to the underside of the lunchbox using a combination of hot glue and superglue.

Once the bamboo stick was firmly attached, I carefully levelled and aligned the motors to ensure that both wheels were positioned correctly and evenly. I then used the hot glue gun to attach the motors to the bamboo stick and further secured them using a zip tie, as shown in the image on the left.

This design was a significant improvement over the previous version because the bamboo support prevents the motors from being pushed further into the body or becoming misaligned. As a result, the wheels remain in a fixed position, making the movement system much more rigid, stable and reliable. This also ensures that the robot's movement is more consistent and reduces the mechanical problems caused by the previous design.

Conclusion:

Overall, the project was successful in demonstrating how a line-following robot can use LDRs, a comparator and transistors to detect and respond to a track. Throughout the build, I was able to apply the theory from my circuit design to a physical prototype and identify several problems that would not have been obvious from the circuit diagram alone.

The sensing circuit worked by using the change in resistance of the LDRs to produce different voltages at TP1 and TP2. These voltages were then compared using the LM393 comparator. This allowed the circuit to determine which side of the robot was moving over the black line and activate the appropriate output. I also had to overcome the limitation of the LM393's open-collector outputs when using the **NPN transistors** I already had available. Although using PNP transistors would have simplified the design, I was able to develop a solution that allowed the NPN transistors to be used successfully.

The mechanical construction presented more challenges than the electronic circuit. Initially, I used a lunchbox as the main body and attached the motors using zip ties and a torch body. However, the motors could move and become misaligned, which affected the robot's movement. The original caster wheel also caused problems because it was loose and resulted in the robot becoming unstable and falling off the track. These issues showed me that the mechanical structure is just as important as the electronic design.

For the second version, I removed the caster wheel and redesigned the motor mounting system. I used a bamboo stick secured to the lunchbox and attached the motors firmly to it using hot glue and zip ties. This prevented the motors from being pushed out of position and made the overall structure much more rigid. The improved design resulted in better wheel alignment and more consistent movement.

Improvements

If I were to develop the robot further, I would make several improvements:

- **Design and manufacture a custom PCB** instead of using a breadboard. This would make the robot more compact, reliable and suitable as a finished product rather than a prototype.
- **Create a custom motor-to-wheel coupler** instead of using superglue. This would allow the wheels to be removed and replaced easily and would provide a more precise connection.
- **Design a proper motor mounting bracket** rather than relying on hot glue and zip ties. A rigid, purpose-built mount would improve the alignment and durability of the robot.
- **Use a more suitable caster wheel or alternative support system** if additional front support is required. It would need to be securely mounted while allowing the robot to turn smoothly.
- **Improve the sensor mounting position** so that the distance between the LDRs and the track remains consistent. This would make the voltage changes more predictable and improve the accuracy of the line detection.
- **Make the overall body more compact and organised**, particularly by reducing the amount of exposed wiring and arranging the components more efficiently.
- **Test and calibrate the LDR thresholds more thoroughly** so that the comparator responds reliably to different lighting conditions and track surfaces.

Overall, the project gave me a much better understanding of the difference between **designing a circuit theoretically and implementing it as a working physical system**. The problems I encountered with the motor alignment, caster wheel and transistor configuration required me to adapt my original designs rather than simply following my first plan. The second version was a significant improvement, particularly in terms of mechanical stability, and the project successfully demonstrated the main principles required for a line-following robot.